

Phase 1 — Leakage Audit

All column-level decisions documented below.

Column	Action	Reason
V138	DROP	>=80% missing in training data
V139	DROP	>=80% missing in training data
V140	DROP	>=80% missing in training data
V141	DROP	>=80% missing in training data
V142	DROP	>=80% missing in training data
V143	DROP	>=80% missing in training data
V144	DROP	>=80% missing in training data
V145	DROP	>=80% missing in training data
V146	DROP	>=80% missing in training data
V147	DROP	>=80% missing in training data
V148	DROP	>=80% missing in training data
V149	DROP	>=80% missing in training data
V150	DROP	>=80% missing in training data
V151	DROP	>=80% missing in training data
V152	DROP	>=80% missing in training data
V153	DROP	>=80% missing in training data
V154	DROP	>=80% missing in training data
V155	DROP	>=80% missing in training data
V156	DROP	>=80% missing in training data
V157	DROP	>=80% missing in training data
V158	DROP	>=80% missing in training data
V159	DROP	>=80% missing in training data
V160	DROP	>=80% missing in training data
V161	DROP	>=80% missing in training data
V162	DROP	>=80% missing in training data
V163	DROP	>=80% missing in training data

Column	Action	Reason
V164	DROP	>=80% missing in training data
V165	DROP	>=80% missing in training data
V166	DROP	>=80% missing in training data
V322	DROP	>=80% missing in training data
V323	DROP	>=80% missing in training data
V324	DROP	>=80% missing in training data
V325	DROP	>=80% missing in training data
V326	DROP	>=80% missing in training data
V327	DROP	>=80% missing in training data
V328	DROP	>=80% missing in training data
V329	DROP	>=80% missing in training data
V330	DROP	>=80% missing in training data
V331	DROP	>=80% missing in training data
V332	DROP	>=80% missing in training data
V333	DROP	>=80% missing in training data
V334	DROP	>=80% missing in training data
V335	DROP	>=80% missing in training data
V336	DROP	>=80% missing in training data
V337	DROP	>=80% missing in training data
V338	DROP	>=80% missing in training data
V339	DROP	>=80% missing in training data
D6	DROP	>80% missing in IEEE-CIS (measured in forensic analysis)
D7	DROP	>80% missing in IEEE-CIS (measured in forensic analysis)
D8	DROP	>80% missing in IEEE-CIS (measured in forensic analysis)
D9	DROP	>80% missing in IEEE-CIS (measured in forensic analysis)
D12	DROP	>80% missing in IEEE-CIS (measured in forensic analysis)
D13	DROP	>80% missing in IEEE-CIS (measured in forensic analysis)
D14	DROP	>80% missing in IEEE-CIS (measured in forensic analysis)
dist2	DROP	93.6% missing (measured in forensic analysis)
TransactionID	DROP	Identifier — not a predictive feature

Column	Action	Reason
isFraud	DROP	Target — must not appear in X
TransactionDT	TRANSFORM	Converted to hour_sin/cos + day_index; raw DT excluded from X to prevent temporal positional memorisation

Key Leakage Prevention Measures

1. **isFraud** excluded from X in all splits.
2. **TransactionID** excluded from X (identifier, not predictive feature).
3. **TransactionDT** converted to hour_sin/cos + day_index; raw excluded from X.
4. All preprocessing (imputation, encoding) fit ONLY on training data.
5. Validation and test data only transformed — never fit.
6. Threshold selected on validation; test evaluated once with frozen threshold.
7. V-columns with >80% missing dropped (measured on training data only).
8. D-columns with >80% missing dropped (D6,D7,D8,D9,D12,D13,D14).
9. dist2 dropped (93.6% missing in training).